TCS

DATA ANALYST INTERVIEW



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Helping you to learn Data Analytics

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Data Analyst Interview–Real-World Scenario Q&A- 2025

Q1. E-commerce Sales Analysis

Scenario: A client wants to know why their total sales are increasing, but profit margins are decreasing. How would you analyze this in Power BI or SQL?

Answer:

I would first break down sales and profit by product category, sub-category, and region. Using Power BI, I'd create measures:

Total Sales = SUM(Sales[SalesAmount])

Total Profit = SUM(Sales[Profit])

Profit Margin = DIVIDE([Total Profit], [Total Sales])

Then, I'd check whether discounts, high freight costs, or low-margin products are driving the drop. A variance analysis (Year-over-Year sales vs. profit margin) would clearly show whether the issue is product mix or pricing strategy.

Q2. Banking – Fraud Detection

Scenario: A bank client wants to detect unusual transactions. What steps will you take as a data analyst?

Answer:

I'd analyze transaction data using SQL to flag anomalies. For example:

- Transactions above a threshold (e.g., ₹5 lakh).
- Multiple transactions in a short time window.
- Transactions outside customer's usual geography.

SQL query example:

```
SELECT CustomerID, TransactionID, Amount, Location, Date
FROM Transactions
WHERE Amount > 500000
   OR Location NOT IN (SELECT HomeLocation FROM Customers WHERE CustomerID = Transactions.CustomerID)
```

Then I'd visualize high-risk customers in Power BI to provide insights for the fraud prevention team.

Q3. Telecom – Customer Churn

Scenario: A telecom client wants to reduce churn. What metrics would you analyze?

Answer:

I'd calculate churn rate using:

ChurnRate=Customers LostTotal Customers at StartChurn Rate = \frac{\text{Customers Lost}}{\text{Total Customers at Start}}

Steps:

1. Identify customers who didn't renew.
2. Segment churn by region, plan type, and usage patterns.
3. Use DAX or SQL to find top reasons (low usage, high charges, poor network coverage).

For example, in Power BI, I'd create a churn dashboard showing:

- Monthly churn trend.
- Churn by plan.
- Customer complaints correlation with churn.

Q4. Healthcare – Hospital Utilization

Scenario: A hospital client wants to optimize bed utilization. What analysis would you perform?

Answer:
I'd calculate:

- Average Length of Stay (ALOS) per department.
- Bed Occupancy Rate = (Occupied Beds / Total Beds) × 100.
- Readmission rates.

SQL query example:

```
SELECT Department,
    AVG(DATEDIFF(day, AdmissionDate, DischargeDate)) AS AvgLengthOfStay,
    COUNT(CASE WHEN DischargeDate IS NULL THEN 1 END) * 100.0 / COUNT(*) AS BedOccupancyRate
FROM Admissions
GROUP BY Department;
```

This would help the hospital identify under-utilized and over-utilized departments.

Q5. Supply Chain – Inventory Optimization

Scenario: A retail client complains about frequent stockouts. How would you solve this?

Answer:
I'd perform a demand-supply gap analysis:

- Compare historical sales with current inventory.
- Forecast demand using moving averages.
- Create a Power BI dashboard with KPIs like Stock-out Rate, Inventory Turnover, and Reorder Level Alerts.

This would help the business maintain optimal stock levels and reduce lost sales.

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Q6. Insurance – Claims Analysis

Scenario: An insurance company wants to detect fraudulent claims. How would you help them?

Answer:
I'd analyze claims data for unusual patterns, e.g.:

- Claims just below approval threshold.
- High frequency of claims from same customer.
- Claims from high-risk regions.

SQL query:

```
SELECT CustomerID, COUNT(ClaimID) AS ClaimCount, AVG(ClaimAmount) AS AvgClaim
FROM Claims
GROUP BY CustomerID
HAVING COUNT(ClaimID) > 5 OR AVG(ClaimAmount) > 100000;
```

This will highlight suspicious customers for further investigation.

Q7. Real Estate – Pricing Model

Scenario: A real estate client wants to understand price variations in different cities. How would you analyze it?

Answer:
I'd clean and analyze property data with fields like location, size, amenities, and price. Steps:

1. Use Power BI to show price per sq. ft by city.
2. Identify top factors influencing price (proximity to metro, hospitals, IT hubs).
3. Compare historical trends to predict price growth.

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